Proactive Social Agency & Epistemic Interception (BESPOKE ENGINE)

Repo: https://github.com/aedmark/bespoke-engine

Roadmap Phase: #033

Subsystem: `src/agency.zig`

Vertical Slice Proof: `proofs/bayou_puzzle.zig` (`--bayou`)

Design Reference: `DECISIONS.MD; D-038`

---

1. Executive Summary

In traditional point-and-click adventure games and RPGs (such as Sierra's SCI engine or LucasArts' SCUMM), non-player characters behave primarily as passive interrogation kiosks. They stand in place or follow fixed patrol paths, waiting for the player to click on them to exhaust dialogue trees. The player holds absolute conversational initiative; NPCs never defend their secrets, react proactively to suspicious behavior, or counter-investigate.

Phase 33 introduces an autonomous social agency subsystem. Wandering actors continuously monitor player action telemetry, evaluate personal agendas, actively hunt down and intercept moving players across spatial graphs, and initiate high-stakes epistemic dialogues (persuasion, dissuasion, warnings, alibi fabrication).

2. The Core Problem in Narrative Game AI

Dimension	Classic Narrative Game (SCI / SCUMM)	BeSpoke Engine (Phase 33)
Initiative	100% Player-driven. NPCs wait to be spoken to.	Shared. NPCs actively seek out and interrupt the player.
Awareness	Global boolean flags (`talked_to_butler = true`).	Relational facts, line-of-sight, and social rumor propagation.
Negative Knowledge	Rare or hardcoded via complex flag branching.	First-class omission queries (e.g. "player has *not* asked about the will").
Spatial Movement	Static loops; characters ignore player location.	Dynamic A* pursuit tracking moving targets across room graphs.
Dialogue Stakes	Exhaustive tree exploration without penalty.	Epistemic contests: lacking evidence leaves the player vulnerable to manipulation.

3. Core Mechanics Explained

A. Action Telemetry & Negative Space (Omissions)

Whenever the player interacts with a scene prop or talks to an NPC, the simulation records a flat relational tuple:

```
(Laura, Investigated, CarriageHouseLock, Permanent)
(Laura, Questioned, Lillian, Permanent)
```

In detective scenarios, what the player has **not** done is often more important than what they have done. An accomplice might attempt to plant a false alibi only *before* the detective examines a crime scene, or exploit the detective's ignorance if a key witness has not been interviewed yet.

The agency engine evaluates omission clauses directly through negation queries against active relational memory:

```
Trigger:
  where Investigated(Laura, Cellar)
    and not Knows(Laura, WilburFootprintMatch)
```

If the condition holds, the agenda transitions from `dormant` to `triggered`.

B. Dynamic Spatial Pursuit

Once triggered, an NPC does not teleport or wait in an adjacent room. The engine assigns the NPC a pursuit motor intent targeting the player:

1. **Continuous Target Tracking:** The agency system samples the player's current waypoint. If the player transitions between rooms or corridors, the pursuer's A* path recalculates periodically or on waypoint entry.
2. **Topology Traversal:** Pursuing NPCs follow identical graph movement rules to the player. They open unlocked doors, walk edge corridors at standard velocities, and cannot pass through locked barriers.

C. The Soft-Interrupt Pattern

In game development, abrupt dialogue cutscenes that freeze actors mid-stride cause severe edge cases: snapped sprite positions, broken pathfinding queues, and camera glitches.

Phase 33 introduces a **soft-interrupt** model:

- The pursuer paths until they reach the same waypoint node as the target player.
- If the player is currently traversing an edge between two waypoints, the NPC yields.
- The instant the player arrives at their destination waypoint and becomes idle, the engine launches the dialogue routine. The player is smoothly rotated to face the pursuer, and the conversational interface engages naturally.

D. Multi-Agent Contention Resolution

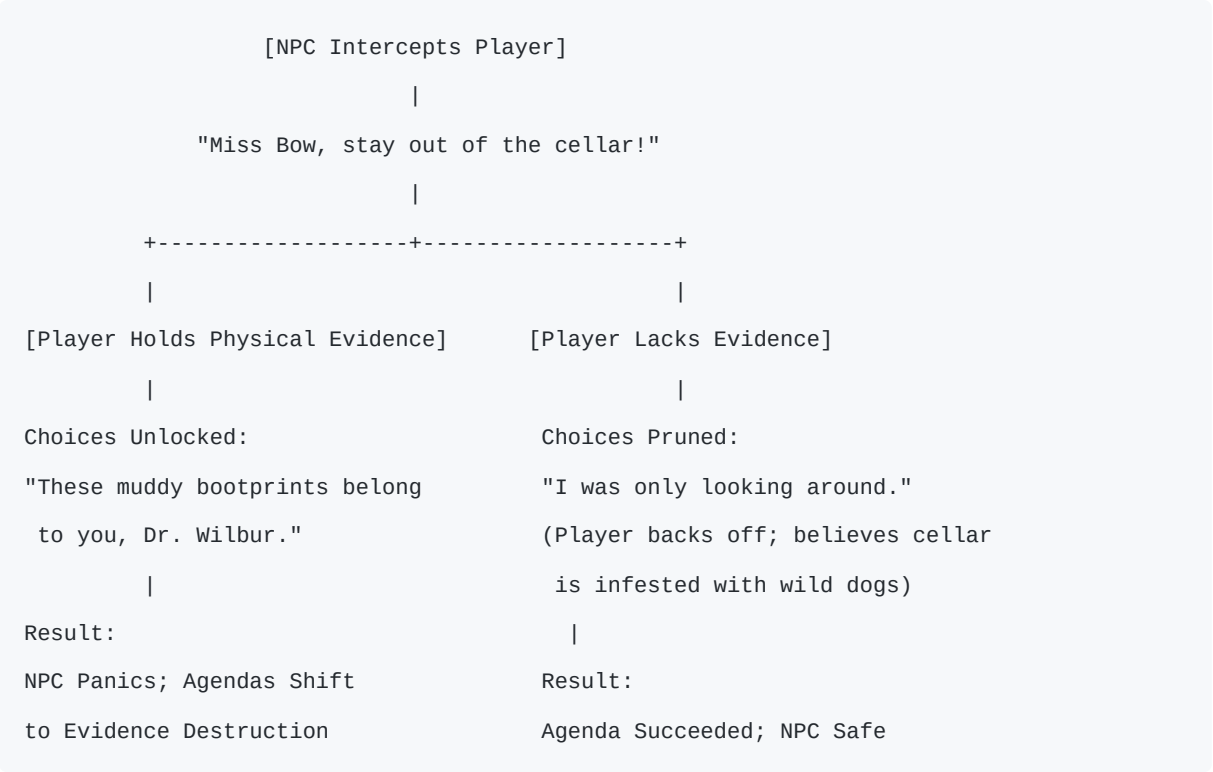
If the player commits an act that alarms multiple suspects simultaneously (for example, breaking into the estate's private study), multiple agendas may trigger at once.

To prevent suspects from dog-piling the player, the subsystem runs a deterministic priority solver:

1. **Priority Ranking:** Each agenda has a base priority rating (e.g. self-preservation > alibi planting > casual gossip).
2. **Proximity Tie-Breaking:** If priorities are identical, the NPC closest to the player along the waypoint graph wins the pursuit lock.
3. **Queueing:** Lower-priority pursuers pause their pursuit or remain in a holding pattern until the active encounter concludes.

E. Epistemic Contests: Dialogue Gated by Evidence

Dialogue encounters initiated by an intercepting NPC are modeled as epistemic contests where truth and belief are dynamic states:



1. **Fact-Guarded Choice Trees:** Dialogue options available to the player are pruned by the engine's query engine. If the player hasn't discovered the physical clue, rebuttal lines are completely hidden.
2. **State Mutations:** The branch chosen writes new relational facts into the world arena (`Believes(Laura, CellarIsDangerous)` or `Panicked(Wilbur)`).
3. **Consequence Cascades:** A panicked suspect abandons their normal daily schedule to hide evidence, lock doors, or flee, altering the downstream simulation.

4. Scenario Walkthrough: "The Bayou Estate"

To validate the implementation, Phase 33 includes a dedicated vertical slice scenario inspired by classic bayou murder mysteries (*proofs/bayou_puzzle.zig*, run with `--bayou`).

The Setting & Characters

- **Location:** A Louisiana plantation house with 12 waypoints (Parlor, Library, Dining Hall, Cellar, Carriage House, Bayou Docks, Cemetery).
- **Laura:** The player character.
- **Dr. Wilbur:** Family physician who poisoned the estate owner; terrified of the carriage house being searched.
- **Ethel:** Heiress seeking to establish a false alibi before the body is discovered.
- **Celie:** Caretaker moving between Kitchen and Docks, sharing overheard rumors.

Scenario Flow Under Test

1. **Exploration:** Laura wanders into the Dining Hall and discovers an altered grandfather clock, recording `Investigated(Laura, DiningClock)`.
2. **Triggering Alibi Inoculation:** Laura heads toward the Parlor. Ethel detects that Laura has not yet spoken to her (`where not Questioned(Laura, Ethel)`). Ethel's agenda triggers: she paths across the corridor, intercepts Laura at the Parlor doorway, and volunteers an unsolicited alibi ("I was reading in the library all afternoon").
3. **The Rebuttal:** Because Laura inspected the dining room clock earlier, an extra dialogue branch appears: *"Then why was the clock in the library stopped at 3:00?"* Ethel's alibi collapses, setting `EthelDiscredited` and causing her to retreat to her bedroom.
4. **Dissuasion Attempt:** Later, Laura approaches the Carriage House. Dr. Wilbur notices. Since Laura does not possess Wilbur's prescription pad (`not Possesses(Laura, PrescriptionPad)`), Wilbur sprints across the courtyard to intercept Laura, warning her of dangerous wild dogs in the marsh. Laura is forced to retreat back to the courtyard unless she can find physical proof of his guilt.

5. Architectural Invariants for Engine Implementers

1. **100% Deterministic Relational Memory:** All agenda states (`dormant` , `pursuing` , `engaged` , `succeeded` , `rebutted`), pursuit targets, and belief mutations exist strictly as flat integer fact tuples in the central memory arena. No state escapes to hidden C-heap allocations.
2. **Zero-Overhead Save/Rewind:** Because agenda state lives in the relational fact arena, saving the game or scrubbing backward on the interactive timeline instantly rewinds pursuit coordinates, cooldowns, and conversational memory with zero custom serialization code.
3. **Decoupled Architecture:** `src/agency.zig` acts as an orchestrator sitting between:
   - `World` (evaluates triggers and writes outcome facts).
   - `SpatialSystem` (requests A* paths and verifies waypoint arrivals).
   - `VM` (launches dialog routines via soft interrupts).
4. **Separation Rule Compliance:** The engine module `src/agency.zig` contains zero narrative names (no "Laura", "Wilbur", or "Bayou"). All triggers, priority values, and target identifiers are passed in as data structures by the scenario host.